25901608 Reg.No:

Name:

MAHATMA GANDHI UNIVERSITY, KOTTAYAM
MGU-UGP (HONOURS) REGULAR/IMPROVEMENT/REAPPEARANCE EXAMINATION APRIL 2026
SECOND SEMESTER
Discipline Specific Core Course (DSC) - MG2DSCBIF100 - BIOLOGICAL DATA AND MANAGEMENT
(2024 ADMISSION ONWARDS)

Duration: 1 Hour 30 Minutes Maximum Marks:50

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A
Short answer Questions
Answer any 5 questions
Each question carries 2 marks

1. Compare cloud databases and on-premises databases in terms of scalability.[U]/[C01]
   Cloud databases scale easily on demand (horizontal/elastic scaling), while on-premises databases have limited scalability that requires manual hardware upgrades.
2. Articulate two features of Medline.[A]/[CO4]
   MEDLINE provides extensive coverage of biomedical and life sciences journals. It uses MeSH (Medical Subject Headings) for precise and standardized searching.
3. Discuss classification of biological database.[U]/[CO2]
   Biological databases are broadly classified into primary databases (store raw experimental data like DNA/protein sequences, e.g., GenBank), secondary databases (contain curated, analyzed, and annotated data, e.g., PROSITE), and composite databases (integrate data from multiple sources, e.g., UniProt).
4. Identify scopes of bilogical data.[U]/[CO2]
   The scope of biological data includes genomic data (DNA/RNA sequences), proteomic data (protein structure and function), metabolomic data (metabolic pathways and compounds), and biodiversity data (species classification and distribution).
5. Give one application of motif databases.[A]/[CO4]
   Motif databases are used to identify conserved sequence patterns in DNA or proteins to predict their function or biological role.
6. Establish why GenBank is considered a primary database.[A]/[CO4]

GenBank is considered a primary database because it stores original, raw nucleotide sequence data submitted directly by researchers without extensive curation or analysis.

7. Write the purpose of BCDB.[A]/[CO4]
   The purpose of BCDB (Bacterial Carbohydrate Structure Database) is to store and provide detailed information on the structures and properties of bacterial carbohydrates for research and analysis.

[2x5 = 10]

Part B
Short Essay Questions
Answer any 4 questions
Each question carries 5 marks

8. Illustrate the flow of data from input to database storage using a suitable diagram and labels.[U]/[C01]
   Diagram representation – 2 marks ,Correct flow/order – 1 mark ,Proper labeling – 1 mark, Explanation– 1 mark
9. Discuss the role of big data in driving decision-making and innovation.[U]/[C01]
   Explanation of big data concept – 1 mark, Role in decision-making – 2 marks, Role in innovation – 1 mark, Clarity with examples/points – 1 mark
10. identify the major databases maintained by NCBI[U]/[CO2]
    NCBI identification – 1 mark, Each correct database (any 3–4 examples) – 3 marks, Clarity and relevance – 1 mark
11. Discuss how primary databases differ from secondary databases?[U]/[CO3]
    Definition of primary databases – 1 mark, Definition of secondary databases – 1 mark, Difference in nature of data (raw vs processed) – 1 mark, Example for primary database – 1 mark, Example for secondary database – 1 mark
12. Articulate the role of the ENZYME database.[A]/[CO4]
    Definition of ENZYME database – 1 mark, Role in enzyme classification (EC numbers) – 1 mark, Function in identifying enzyme activity – 1 mark, Use in biochemical research/pathways – 1 mark, Clarity – 1 mark
13. Complete a table listing major genome databases.[A]/[CO4]
    Correct table format – 1 mark, At least 3–4 correct databases – 2 marks, Appropriate descriptions – 1 mark, Clarity and presentation – 1 mark

[5x4 = 20]

Part C

Long essay Questions

Answer any 2 questions.

Each question carries 10 marks

14. Compare the main features of SQL, PL/SQL, and T-SQL.[U]/[CO1]
    Definition of SQL – 2 marks, Definition of PL/SQL – 2 marks, Definition of T-SQL – 2 marks, Comparison of features (any 3–4 points) – 3 marks, Clarity and presentation – 1 mark.
15. Compare and contrast the three-tier architecture of a DBMS with the two-tier architecture.[U]/[CO1]
    Explanation of two-tier architecture – 2 marks, Explanation of three-tier architecture – 2 marks, Structural comparison – 2 marks, Functional differences (processing, scalability, security) – 3 marks, Clarity– 1 mark
16. Describe the key phases in the growth of biological databases.[U]/[CO2]
    Early phase explanation – 2 marks, Primary database phase – 2 marks, Specialized database phase – 2 marks, Integration phase – 2 marks, Big data phase – 2 marks
17. Explain why structural databases are useful.[A]/[CO4]
    Definition of structural databases – 2 marks, Importance in understanding structure-function relationship – 2 marks, Role in drug design – 2 marks, Role in studying molecular interactions/mechanisms – 2 marks, Example and clarity of explanation – 2 marks

[10x2 = 20]